

# VidaReal Translator

Automatic AI translation of the Rock RMS interface with permanent caching — built in-house, without modifying Rock's core.

Executive brief · September 2026 · Plugin version 1.4.6

## 1 Executive summary

Rock RMS — the platform that manages the people, groups, events and check-in of the church — ships **in English only**. That slows adoption among staff and volunteers. **VidaReal Translator** translates the entire interface into Spanish (or any language) automatically and transparently for the user.

**The core idea:** every interface string is translated by AI **once** and is then stored in our own database. From the second user onward, the translation comes out of the cache — instant and free. AI spend trends to zero as usage grows.

### Once

each string's translation is paid for a single time; after that it is permanently cached

### Zero changes

to Rock's code: a standalone plugin, updatable and reversible

### Interface only

layered filters keep member data out of the AI; the exclusion rules are audited on every Rock upgrade

### One click

to switch it on, or remove it from every Rock site at once

## 2 What the user sees

- **The whole interface in Spanish:** menus, buttons, forms, validation messages and help text — on the intranet, the public site and check-in.
- **Optional language switcher** (Weglot-style): a discreet tab to switch between Spanish, English or other languages. The preference is remembered per device.
- **Zero friction:** nothing to install, no change to their workflow. If anything fails, the page simply shows in English — it never breaks.

### 3 Architecture

Four pieces, each with one clear responsibility. The numbered arrows follow the journey of a new string:

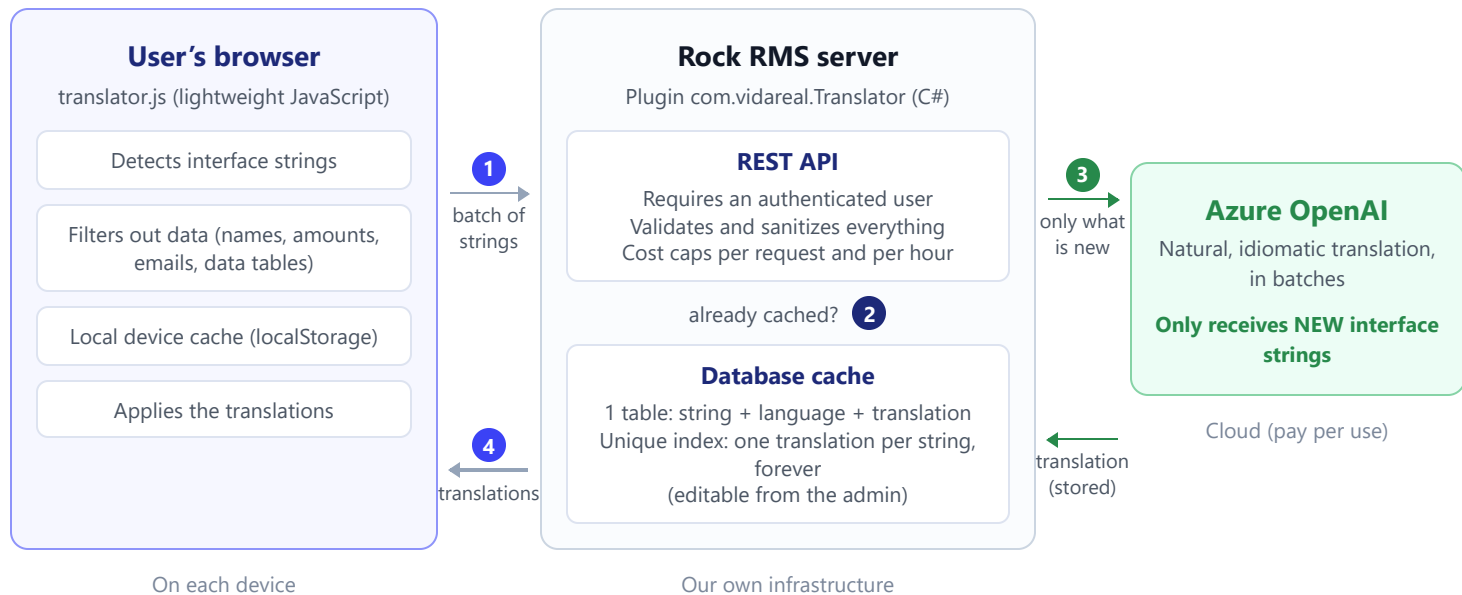


Fig. 1 — Architecture. The browser never talks to Azure directly; everything goes through the Rock server, which validates, applies caps and caches.

### Components

| Component                      | Responsibility                                                                                                                                                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>translator.js (browser)</b> | Detects the visible interface strings live — including classic screens (WebForms), modern ones (Obsidian/Vue) and AJAX-loaded content — and applies the translations without reloading. Includes the language switcher. |
| <b>REST API (C# plugin)</b>    | Receives batches of strings, resolves them against the database cache and only sends the AI what has never been translated. Requires an authenticated user; sanitizes the AI output; enforces cost limits.              |
| <b>Database</b>                | A single translations table with a unique index on (string, language). It is the permanent memory of the system and is managed from within Rock.                                                                        |
| <b>Azure OpenAI</b>            | Translates software vocabulary (buttons, menus, messages) with human-level quality, under strict instructions: never touch proper names, codes or variables.                                                            |

## 4 How it works at the DOM level

The translator touches no server-side HTML or templates: it operates on the **already-rendered DOM** in the browser, as a visual layer. Every string goes through a 5-step pipeline:

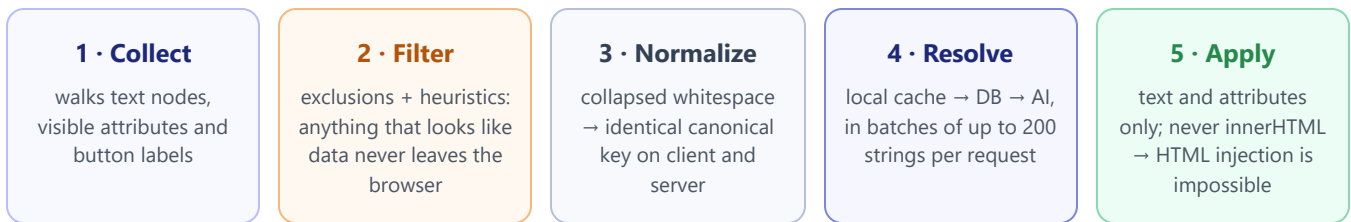


Fig. 2 — Pipeline in the browser. Steps 1–3 and 5 are local; only step 4 can touch the network.

### Anatomy of a page: what it touches and what it does not

```
<label>First Name</label> ✓ translates interface text
<input value="Juan Pérez"> × untouched form value: whatever is typed or saved is never touched
<input type="submit" value="Save"> ✓ translates here value is the visible button label, not data
<div title="Print this page"> ✓ translates visible attributes: title, placeholder, aria-label, alt
<table class="grid-table">
  <th>Email Address</th> ✓ translates column header = interface
  <td>juan@vidareal.tv</td> × untouched data cell: excluded wholesale, never even sent
<span>{{ Person.NickName }}</span> × untouched Lava token / variable
<textarea>...</textarea>, <pre>, <code>, contenteditable × untouched editable or technical content,
excluded wholesale
```

In addition, any element can be excluded individually with the class **nottranslate** or the attribute **data-no-translate**, and there are selector lists configurable from the admin without recompiling.

- **Idempotent and efficient:** nodes that have already been translated are marked and never reprocessed; strings already known come out of the cache without touching the network.
- **Safe application:** text is replaced node by node, preserving edge whitespace; innerHTML is never used, so the AI cannot introduce HTML or scripts.
- **Fail-safe:** every step runs inside error guards; an invalid selector or an odd response is discarded without affecting the page.

### Coverage of dynamic content

Rock builds its pages from blocks that arrive at different moments. The translator covers all four paths:

| Scenario                                            | Mechanism                                                                                                                                                     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial load and late render                        | Staggered passes (0 / 600 / 1500 ms) over the whole page.                                                                                                     |
| Content injected by AJAX / modern blocks (Obsidian) | <b>MutationObserver</b> : watches the DOM and processes <i>only the new subtrees</i> (it does not re-sweep the page), grouping changes with a small debounce. |
| Partial postbacks (classic WebForms screens)        | Hook into the ASP.NET AJAX lifecycle (PageRequestManager): re-scans when each postback finishes.                                                              |
| Navigation without reload (SPA)                     | Hook into the History API (pushState / popstate): re-scans on route change.                                                                                   |

### 5 The cache funnel: why the cost trends to zero

Every string on the page passes through three levels. Only the last one costs money, and each string crosses it **once in the lifetime of the system**:

100% of the strings on the page

every visit



With use, levels 1 and 2 absorb practically everything: the Rock interface is finite, and once it is translated, level 3 is barely visited.

Fig. 3 — Resolution funnel. Cost lives only at level 3, and each string pays it exactly once.

### Built-in cost controls

| Control                           | Effect                                                                                                                                                                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Permanent cache (DB + browser)    | Each string is paid for once in the lifetime of the system.                                                                                                                                                             |
| Per-request and hourly limits     | A maximum number of strings per request and a global cap on new translations per hour: an uncontrolled spending spike is impossible.                                                                                    |
| Deduplication and anti-repetition | Strings repeated on the page and simultaneous requests are consolidated: the same thing is never paid for twice.                                                                                                        |
| No credentials → no spend         | If Azure is not configured, the plugin simply does not translate. It never generates accidental spend.                                                                                                                  |
| Global hourly cap                 | The public sites must translate for visitors without a login, so the service is reachable without a Rock session. The defence against abuse is therefore the hourly cap on <i>new</i> translations, not authentication. |

## 6 Protection of member data

A non-negotiable design rule: «**interface yes, data no**». The filter runs in the browser *before* anything is sent: whatever looks like data never leaves the device.

| ✓ Translated                                                        | × Never sent, never touched                                                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Buttons, menus, titles, form labels, validation messages, help text | People's names, emails, phone numbers, amounts, dates, the contents of data tables, fields that are typed in or saved, editable content |

- The values that travel to the server are excluded by design: a translation **cannot corrupt** what gets stored in the database.
- The AI response is sanitized (any code or markup is stripped) before being stored or displayed.
- Aligned with the Responsible AI Use Policy: only software vocabulary is meant to reach the AI. The filter recognises data by how Rock *renders* it, so a Rock upgrade that changes the markup can open a gap — as happened in September 2026, when a redesigned data grid stopped matching the rule.
- **Keeping the rule true:** exclusions are re-verified on every Rock upgrade; the cache is audited periodically (a proper name in it means a rule has a gap); and self-changing text (counters, clocks) is blocked generically.

## 7 Administration (inside Rock itself)

Everything is managed from a single admin page: *Admin Tools* → *Installed Plugins* → *VidaReal Translator*. No need to touch servers or databases.

| Function                 | Description                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>On / off</b>          | A single toggle activates the translator on <b>every Rock site at once</b> ; switching it off removes it just as cleanly. Fully reversible.                                                                                |
| <b>Editorial control</b> | A listing lets you search any translation and <b>correct it by hand</b> (the correction is locked in: the AI will not touch it again), <b>exclude</b> strings that must never be translated, or delete them to regenerate. |
| <b>Cache purge</b>       | Regenerates every translation (e.g. after improving the AI style instructions). Administrators only.                                                                                                                       |
| <b>Configuration</b>     | Target language, Azure credentials (stored <b>encrypted</b> ), the languages in the switcher and fine-grained scope settings — all through the standard Rock form.                                                         |

## 8 Quality and robustness

- **Graceful degradation:** in the face of any failure (AI down, network, configuration), the page is shown in English. The translator never breaks a screen or a save.
- **Professional translation:** the AI is instructed to use standard software terminology in Spanish, a consistent tone and a similar length (so the layout does not break).
- **Verified integrity:** automated tests cover the data safeguards and the exact parity between the browser code and the server code.
- **Built to grow:** it supports several languages in parallel (each with its own cache) and the architecture allows switching AI provider without rewriting the system.

## In summary

**A one-time development investment** that makes Rock accessible in Spanish for the whole team, with a marginal and decreasing operating cost (trending to zero), full editorial control and zero risk to the data of the congregation.

The same architecture works for any target language: each one keeps its own permanent cache, and adding one does not change a single line of Rock.